

LINUX



Essentials for Clouds





Linux

Family of open-source Unix-like operating systems based on the Linux kernel, an operating system. kernel first released on September 17, 1991, by Linus Torvalds. Linux is typically packaged in a Linux distribution.

Linux is one of the most prominent examples of free and open-source software collaboration. The source code may be used, modified and distributed commercially or non-commercially by anyone under the terms of its respective licenses, such as the GNU General Public License.

90% of all cloud infrastructure is powered by Linux including supercomputers and cloud providers.

74% of smartphones in the world are Linux-based.



Bootloader

A bootloader, also spelled as boot loader or called boot manager and bootstrap loader, is a computer program that is responsible for booting a computer.

When a computer is turned off, its software—including operating systems, application code, and data—remains stored on non-volatile memory. When the computer is powered on, it typically does not have an operating system or its loader in random-access memory (RAM). The computer first executes a relatively small program stored in read-only memory (ROM, and later EEPROM, NOR flash) along with some needed data, to initialize RAM (especially on x86 systems), to access the nonvolatile device (usually block device, eg NAND flash) or devices from which the operating system programs and data can be loaded into RAM.

For example GNU GRUB, LILO, SYSLINUX, or Gummiboot. This is a program that loads the Linux kernel into the computer's main memory, by being executed by the computer when it is turned on and after the firmware initialization is performed.



Kernel

Kernel is central component of an operating system that manages operations of computer and hardware. It basically manages operations of memory and CPU time. It is core component of an operating system. Kernel acts as a bridge between applications and data processing performed at hardware level using inter-process communication and system calls.

Kernel loads first into memory when an operating system is loaded and remains into memory until operating system is shut down again. It is responsible for various tasks such as disk management, task management, and memory management. It decides which process should be allocated to processor to execute and which process should be kept in main memory to execute. It basically acts as an interface between user applications and hardware. The major aim of kernel is to manage communication between software i.e. user-level, applications and hardware i.e., CPU and disk memory



Shell

shell is a computer program which exposes an operating system's services to a human user or other program. In general, operating system shells use either a command-line interface (CLI) or graphical user interface (GUI), depending on a computer's role and particular operation. It is named a shell because it is the outermost layer around the operating system. Command-line shells require the user to be familiar with commands and their calling syntax, and to understand concepts about the shell-specific scripting language (for example, bash).

Graphical shells place a low burden on beginning computer users, and are characterized as being easy to use. Since they also come with certain disadvantages, most GUI-enabled operating systems also provide CLI shells.



Linux Shell

The shell can be defined as a command interpreter within an operating system like Linux/GNU or Unix. It is a program that runs other programs. The shell facilitates every user of the computer as an interface to the Unix/GNU Linux system. Hence, the user can execute different tools/utilities or commands with a few input data.

The shell sends the result to the user over the screen when it has completed running a program which is the common output device. That's why it is known as "command interpreter".

The shell is not just a command interpreter. Also, the shell is a programming language with complete constructs of a programming language such as functions, variables, loops, conditional execution, and many others.

For this reason, GNU/Unix Linux Shell is stronger than the Windows shell.



File System

File system is a method of organizing and retrieving files from a storage medium (e.g., hard drive).

File systems usually consist of files separated into groups called directories. Directories can contain files or additional directories. Today, the most commonly used file system with Windows is NTFS.

Without file management, all files would have no organization and it would be impossible for a file with the same name to exist. Files are most often managed in a hierarchy, which allows you to view files in the current directory and then navigate into any subdirectories.

eg. FAT, FAT32, NTFS, EXT2, EXT3, EXT4



Ext4 File System

The EXT4 filesystem primarily improves performance, reliability, and capacity. To improve reliability, metadata and journal checksums were added. To meet various mission-critical requirements, the filesystem timestamps were improved with the addition of intervals down to nanoseconds. The addition of two high-order bits in the timestamp field defers the Year 2038 problem until 2446—for EXT4 filesystems, at least.

In EXT4, data allocation was changed from fixed blocks to extents. An extent is described by its starting and ending place on the hard drive. This makes it possible to describe very long, physically contiguous files in a single inode pointer entry, which can significantly reduce the number of pointers required to describe the location of all the data in larger files. Other allocation strategies have been implemented in EXT4 to further reduce fragmentation.



Linux Swap Partition

The swap partition serves as overflow space for your RAM. If your RAM fills up completely, any additional applications will run off the swap partition rather than RAM.

This may sound like an easy way to increase your computer's amount of usable memory without actually getting more RAM, but that isn't the case. RAM is the ideal hardware for memory because it's extremely quick, unlike hard drives which are, relatively speaking, much slower.

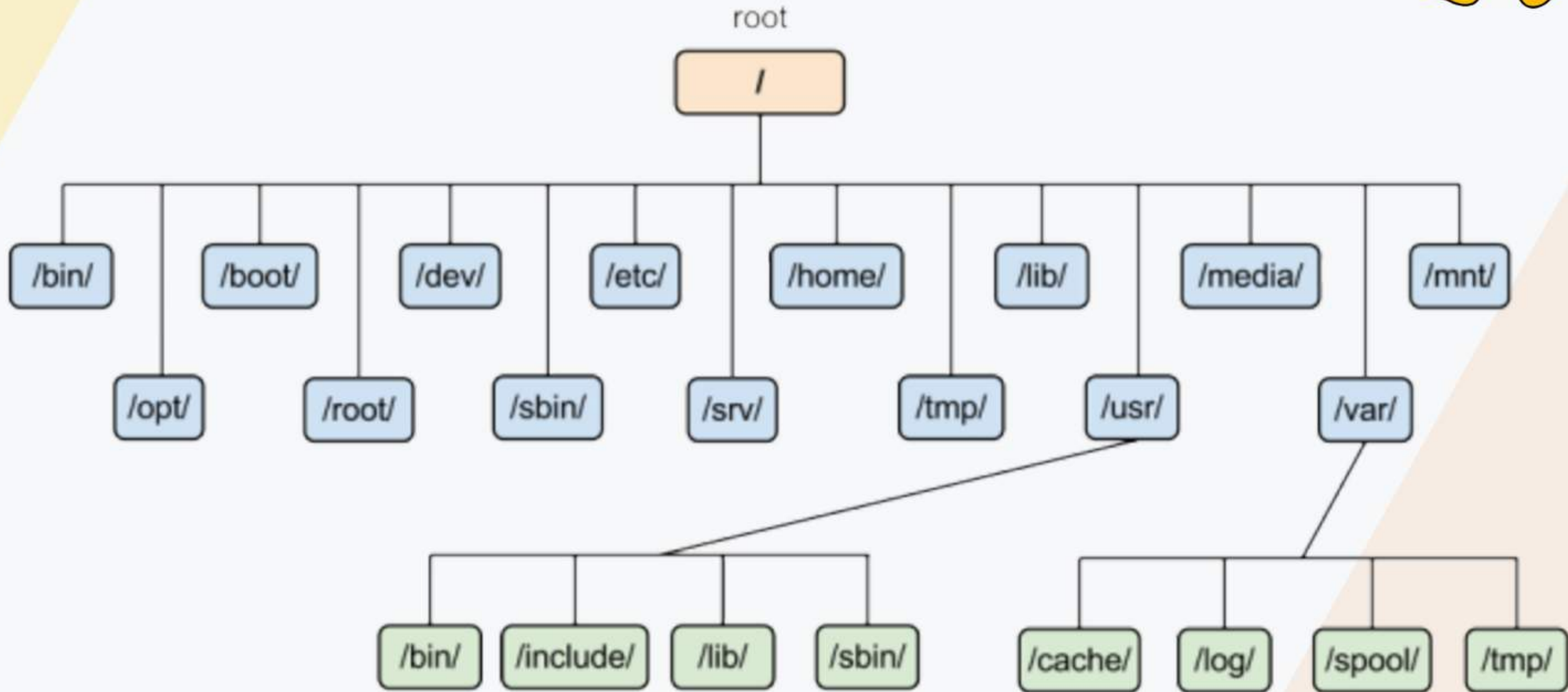
The Linux swap partition is not limited to being overflow storage space. It can assist your PC in other ways.

Best Size = $2 * \text{RAM size}$

Smallest = $\text{Sqrt}(\text{RAM size})$



Linux Structure



Linux Structure



Each of the above directory (which is a file, at the first place) contains important information, required for booting to device drivers, configuration files, etc. Describing briefly the purpose of each directory, we are starting hierarchically.

- **/bin : All the executable binary programs (file) required during booting, repairing, files required to run into single-user-mode, and other important, basic commands viz., cat, du, df, tar, rpm, wc, history, etc.**
- **/boot : Holds important files during boot-up process, including Linux Kernel.**
- **/dev : Contains device files for all the hardware devices on the machine e.g., cdrom, cpu, etc**
- **/etc : Contains Application's configuration files, startup, shutdown, start, stop script for every individual program.**
- **/home : Home directory of the users. Every time a new user is created, a directory in the name of user is created within home directory which contains other directories like Desktop, Downloads, Documents, etc.**
-



Linux Structure

- **/lib** : The Lib directory contains kernel modules and shared library images required to boot the system and run commands in root file system.
- **/lost+found** : This Directory is installed during installation of Linux, useful for recovering files which may be broken due to unexpected shut-down.
- **/media** : Temporary mount directory is created for removable devices viz., media/cdrom.
- **/mnt** : Temporary mount directory for mounting file system.
- **/opt** : Optional is abbreviated as opt. Contains third party application software. Viz., Java, etc.
- **/proc** : A virtual and pseudo file-system which contains information about running process with a particular Process-id aka pid.
- **/root** : This is the home directory of root user and should never be confused with ‘/’
- **/run** : This directory is the only clean solution for early-runtime-dir problem.
- **/sbin** : Contains binary executable programs, required by System Administrator, for Maintenance. Viz., iptables, fdisk, ifconfig, swapon, reboot, etc.

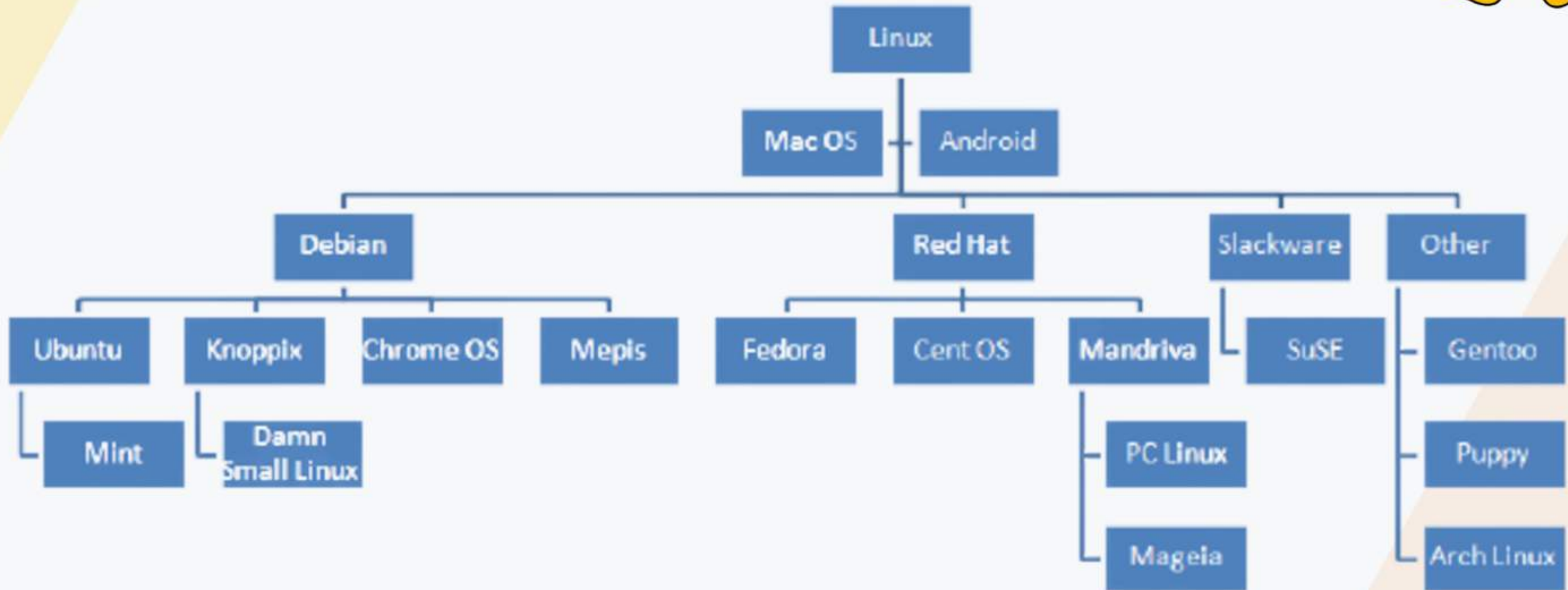
Linux Structure



- **/sbin** : Contains binary executable programs, required by System Administrator, for Maintenance. Viz., iptables, fdisk, ifconfig, swapon, reboot, etc.
- **/srv** : Service is abbreviated as 'srv'. This directory contains server specific and service related files.
- **/sys** : Modern Linux distributions include a /sys directory as a virtual filesystem, which stores and allows modification of the devices connected to the system.
- **/tmp** : System's Temporary Directory, Accessible by users and root. Stores temporary files for user and system, till next boot.
- **/usr** : Contains executable binaries, documentation, source code, libraries for second level program.
- **/var** : Stands for variable. The contents of this file is expected to grow. This directory contains log, lock, spool, mail and temp files.



Linux families





Different linux Distribution

Redhat Based :

Redhat
fedora
openSUSE
centOS ..
and many more

Debian Based :

Ubuntu
Debian
Astra
BOSS
Bodhi
Garuda ..
and many more



UBUNTU

Ubuntu is a Linux distribution based on Debian and composed mostly of free and open-source software. Ubuntu is officially released in three editions: Desktop, Server, and Core for Internet of things device and robots. All the editions can run on the computer alone, or in a virtual machine. Ubuntu is a popular operating system for cloud computing, with support for OpenStack. Ubuntu's default desktop has been GNOME, since version 17.10.

Ubuntu is released every six months, with long-term support (LTS) releases every two years.

Ubuntu is developed by Canonical and a community of other developers, under a meritocratic governance model. Canonical provides security updates and support for each Ubuntu release, starting from the release date and until the release reaches its designated end-of-life (EOL) date.



Linux Terminal

Linux terminal is a standard way of interacting with the Linux server.

The application of the command line might look like a difficult task but it's very convenient if we begin with the basics and create our skills from there.

The terminal emulator can be defined as a program that permits the terminal to use it inside a graphical environment. Most people use the operating system along with a GUI for their daily computer requirements, so terminal emulator usage is essential for most of the users of the Linux server. Users can efficiently and quickly interact because the text is light over resources. All the actions the users can implement by this textual interface. Users will connect to an individual master program for coordinating different program's execution.

The user award was wrapped in the shell programs. Users can write the scripts of share that automated the long series of shell commands for making difficult tasks easier.



Basic Commands

- `uname (a)`
- `hostname (l)`
- `pwd`
- `who`
- `whoami`
- `id`
- `cal`
- `date`
- `time`
- `uptime`
- `w`
- `lscpu`
- `clear`
- `echo "..."`
- `finger user`
- `touch`
- `gedit`
- `mkdir`
- `cp (r/R)`
- `mv`
- `cd`
- `ls (a/al)`
- `locate`
- `cat`
- `man`
- `rm (rf)`
- `rmdir`
- `head/tail`
- `wc`
- `more/less`
- `fdisk`
- `free`
- `ps (a/al)`
- `top`
- `bg/fg`
- `kill`
- `jobs`
- `df`
- `du`
- `mount/umount`
- `zip/unzip`
- `unar`
- `tar (cvf/xvf)`
- `su`
- `sudo su`
- `sudo -i`
- `chmod`
- `chown`
- `systemctl`
- `init 0,6`
- `vi/vim/nano`
- `cat /proc/cpuinfo`
- `cat /proc/meminfo`
- `useradd/userdel`
- `netstat`
- `reboot [min]`
- `shutdown [min]`
- `ip addr show`
- `ifconfig`
- `ping`
- `wget <file.link>`
- `grep <word> <file.txt>`
- `curl <web-address>`
- `tree`
- `whois domain`
- `dig domain`
- `dig -x host`
- `nmtui`
- `nmcli`



Root User

root is the user name or account that by default has access to all commands and files on a Linux or other Unix-like operating system. It is also referred to as the root account, root user and the superuser.

The word root also has several additional, related meanings when used as part of other terms, and thus it can be a source of confusion to people new to Unix-like systems.

One of these is the root directory, which is the top level directory on a system. That is, it is the directory in which all other directories, including their sub directories, and files reside. The root directory is designated by a forward slash (/).

to switch to root

sudo su

sudo -i



Root privileges

Root privileges are the powers that the root account has on the system. The root account is the most privileged on the system and has absolute power over it (i.e., complete access to all files and commands). Among root's powers are the ability to modify the system in any way desired and to grant and revoke access permissions (i.e., the ability to read, modify and execute specific files and directories) for other users, including any of those that are by default reserved for root.

The use of the term root for the all-powerful administrative user may have arisen from the fact that root is the only account having write permissions (i.e., permission to modify files) in the root directory. The root directory, in turn, takes its name from the fact that the filesystems (i.e., the entire hierarchy of directories that is used to organize files) in Unix-like operating systems have been designed with a tree-like (although inverted) structure in which all directories branch off from a single directory that is analogous to the root of a tree.



Package

A package is usually referred to an application but it could be a GUI application, command line tool or a software library (required by other software programs). A package is essentially an archive file containing the binary executable, configuration file and sometimes information about the dependencies.

Linux distributions created their own packaging format to provide the end users ready-to-use binary files (precompiled software) for installing software along with some metadata (version number, description) and dependencies.

eg : a media player (vlc)

a compiler (gcc)

or a simple web server (apache)



Package Manager

A package manager is a tool that allows users to install, remove, upgrade, configure and manage software packages on an operating system. The package manager can be a graphical application like a software center or a command line tool like apt-get or pacman.

Eg : RPM, YUM, APT, DPKG, Snap etc



APT Package Manager

Advanced Package Tool, more commonly known as APT, is a package management system for Debian, Ubuntu, and other similar Linux distributions. It acts as a front-end to the lower-level dpkg package manager, which is used for installing, managing, and providing information on .deb packages. In addition to these functions, APT interfaces with repositories to obtain packages and also provides very efficient dependency management.

eg : `sudo apt install [package]`

`sudo apt update`

`sudo apt upgrade`



IP Address

An IP address represents a unique address that distinguishes any device on the internet or any network from another. IP or Internet Protocol defines the set of commands directing the setup of data transferred through the internet or any other local network.

An IP address is the identifier that enables your device to send or receive data packets across the internet. It holds information related to your location and therefore making devices available for two-way communication. The internet requires a process to distinguish between different networks, routers, and websites. Therefore, IP addresses provide the mechanism of doing so, and it forms an indispensable part in the working of the internet. You will notice that most of the IP addresses are essentially numerical. Still, as the world is witnessing a colossal growth of network users, the network developers had to add letters and some addresses as internet usage grows



Private IP addresses

All the devices that are linked with your internet network are allocated a private IP address. It holds computers, desktops, laptops, smartphones, tablets, or even Wi-Fi-enabled gadgets such as speakers, printers, or smart Televisions. With the expansion of IoT (internet of things), the demand for private IP addresses at individual homes is also seemingly growing. However, the router requires a method to identify these things distinctly. Therefore, your router produces unique private IP addresses that act as an identifier for every device using your internet network. Thus, differentiating them from one another on the network.



Public IP addresses

A public IP address or primary address represents the whole network of devices associated with it. Every device included within with your primary address contains their own private IP address. ISP is responsible to provide your public IP address to your router. Typically, ISPs contains the bulk stock of IP addresses that they dispense to their clients. Your public IP address is practiced by every device to identify your network that is residing outside your internet network.



Dynamic IP addresses

As the name suggests, Dynamic IP addresses change automatically and frequently. With this types of IP address, ISPs already purchase a bulk stock of IP addresses and allocate them in some order to their customers. Periodically, they re-allocate the IP addresses and place the used ones back into the IP addresses pool so they can be used later for another client. The foundation for this method is to make cost savings profits for the ISP.



Static IP addresses

In comparison to dynamic IP addresses, static addresses are constant in nature. The network assigns the IP address to the device only once and, it remains consistent. Though most firms or individuals do not prefer to have a static IP address, it is essential to have a static IP address for an organization that wants to host its network server. It protects websites and email addresses linked with it with a constant IP address.

IPv4



IP stands for Internet Protocol and v4 stands for Version Four (IPv4).

IP version four addresses are 32-bit integers which will be expressed in decimal notation.

Example- 192.0.2.126 could be an IPv4 address.

Parts of IPv4

- ***Network part:*** The network part indicates the distinctive variety that's appointed to the network. The network part conjointly identifies the category of the network that's assigned.
- ***Host Part:*** The host part uniquely identifies the machine on your network. This part of the IPv4 address is assigned to every host. For each host on the network, the network part is the same, however, the host half must vary.
- ***Subnet number:*** This is the nonobligatory part of IPv4. Local networks that have massive numbers of hosts are divided into subnets and subnet numbers are appointed to that.



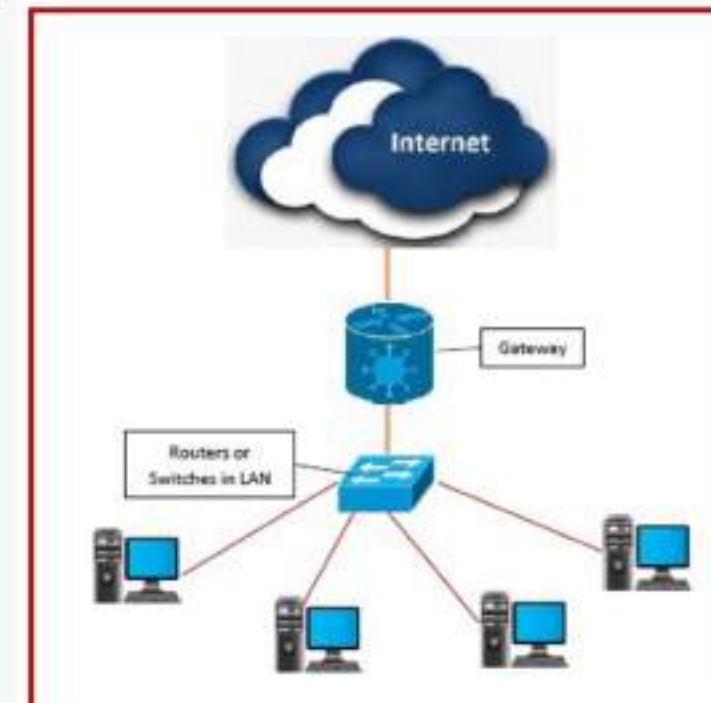
Subnet Mask

Subnet mask is a 32-bit number that masks an IP address and divides the IP address into network address and host address. Subnet Mask is made by setting network bits to all "1"s and setting host bits to all "0"s. Within a given network, two host addresses are reserved for a special purpose, and cannot be assigned to hosts. The "0" address is assigned a network address and "255" is assigned to a broadcast address, and they cannot be assigned to hosts.

Gateways



A gateway is a network node that forms a passage between two networks operating with different transmission protocols. The most common type of gateways, the network gateway operates at layer 3, i.e. network layer of the OSI (open systems interconnection) model. However, depending upon the functionality, a gateway can operate at any of the seven layers of OSI model. It acts as the entry – exit point for a network since all traffic that flows across the networks should pass through the gateway. Only the internal traffic between the nodes of a LAN does not pass through the gateway.





Firewall

The Linux kernel includes the Netfilter subsystem, which is used to manipulate or decide the fate of network traffic headed into or through your server. All modern Linux firewall solutions use this system for packet filtering.

The kernel's packet filtering system would be of little use to administrators without a userspace interface to manage it. This is the purpose of iptables: When a packet reaches your server, it will be handed off to the Netfilter subsystem for acceptance, manipulation, or rejection based on the rules supplied to it from userspace via iptables. Thus, iptables is all you need to manage your firewall, if you're familiar with it, but many frontends are available to simplify the task.



Ubuntu firewall Rules

- **sudo ufw enable**
- **sudo ufw allow 22**
- **sudo ufw status**
- **sudo ufw status verbose**
- **sudo ufw app list**
- **sudo ufw status numbered**
- **sudo ufw disable && sudo ufw enable**
- **sudo ufw deny 22**
- **sudo ufw delete deny 22**
- **sudo ufw reset**



SSH Server

Secure Shell (SSH) is a cryptographic network protocol for operating network services securely over an unsecured network. Typical applications include remote command-line, login, and remote command execution, but any network service can be secured with SSH.

SSH provides a secure channel over an unsecured network by using a client-server architecture, connecting an SSH client application with an SSH server. The protocol specification distinguishes between two major versions, referred to as SSH-1 and SSH-2. The standard TCP port for SSH is 22. SSH is generally used to access Unix-like operating systems, but it can also be used on Microsoft Windows. Windows 10 uses OpenSSH as its default SSH client and SSH server. install openssh-server

apt install openssh-client

apt update

ssh username@ip

ssh username@<host>

Apache Server



The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards. The Apache HTTP Server ("httpd") was launched in 1995 and it has been the most popular web server on the Internet since April 1996

Apache is a popular open-source, cross-platform web server that is, by the numbers, the most popular web server in existence. It's actively maintained by the Apache Software Foundation.

Some high-profile companies using Apache include Cisco, IBM, Salesforce, General Electric, Adobe, VMware, Xerox, LinkedIn, Facebook, Hewlett-Packard, AT&T, Siemens, eBay, and many more installation:

- **apt install apache2**
- **apt update**
- **#port no : 80, 443**
- **ufw allow 80**
- **systemctl restart apache**

Vi Editor



The vi editor is elaborated as visual editor. It is installed in every Unix system. In other words, it is available in all Linux distros. It is user-friendly and works same on different distros and platforms. It is a very powerful application. An improved version of vi editor is vim.

The vi editor has two modes:

- ***Command Mode:*** In command mode, actions are taken on the file. The vi editor starts in command mode. Here, the typed words will act as commands in vi editor. To pass a command, you need to be in command mode.
- ***Insert Mode:*** In insert mode, entered text will be inserted into the file. The Esc key will take you to the command mode from insert mode.

By default, the vi editor starts in command mode. To enter text, you have to be in insert mode, just type 'i' and you'll be in insert mode. Although, after typing i nothing will appear on the screen but you'll be in insert mode. Now you can type anything.

To exit from insert mode press Esc key, you'll be directed to command mode.

Dynamic Host Configuration Protocol



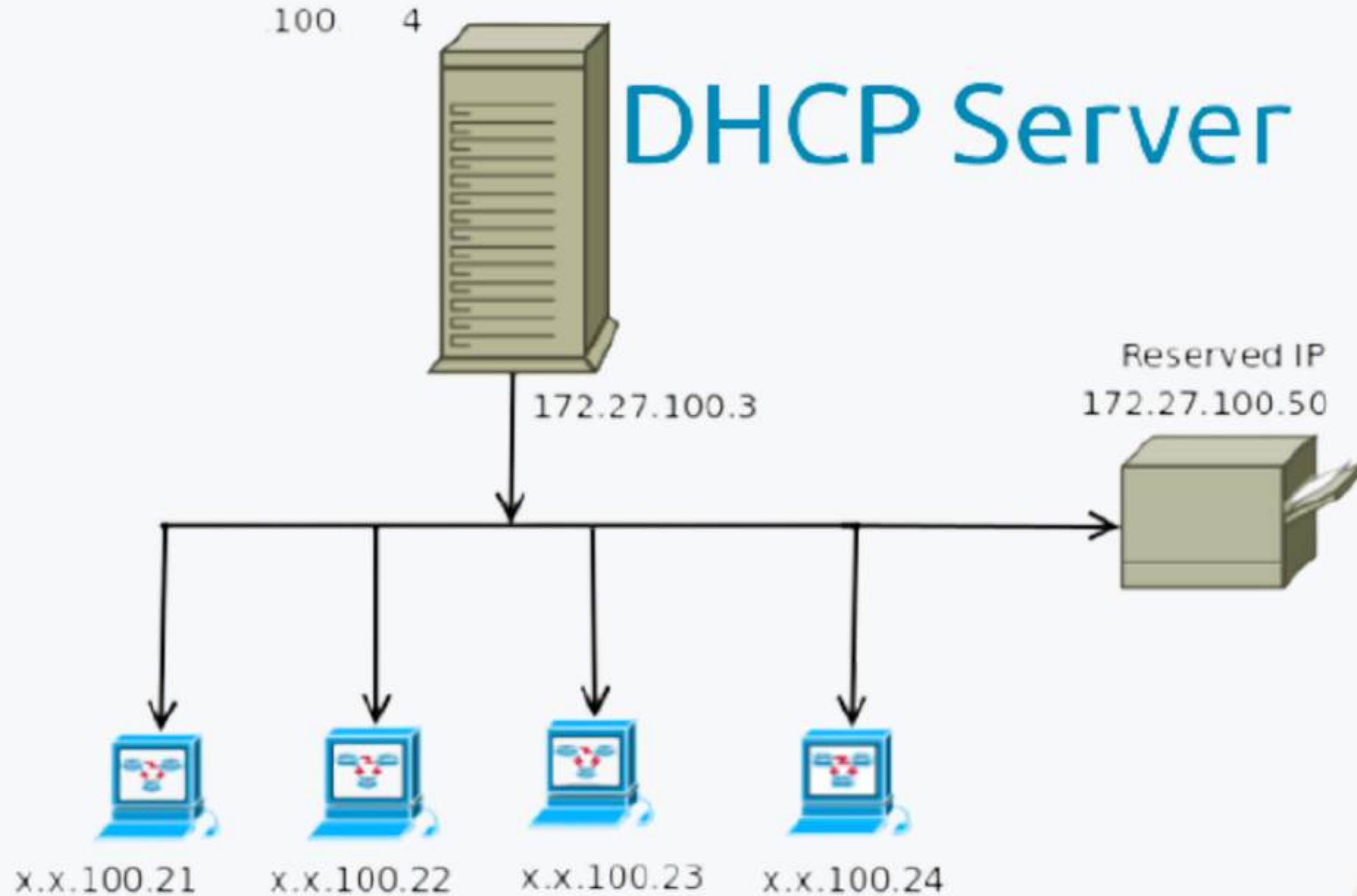
The Dynamic Host Configuration Protocol (DHCP) is a network management protocol used on Internet Protocol (IP) networks for automatically assigning IP addresses and other communication parameters to devices connected to the network using a client–server architecture.[1]

The technology eliminates the need for individually configuring network devices manually, and consists of two network components, a centrally installed network DHCP server and client instances of the protocol stack on each computer or device. When connected to the network, and periodically thereafter, a client requests a set of parameters from the DHCP server using the DHCP protocol.

DHCP can be implemented on networks ranging in size from residential networks to large campus networks and regional ISP networks.[2] Many routers and residential gateways have DHCP server capability. Most residential network routers receive a unique IP address within the ISP network. Within a local network, a DHCP server assigns a local IP address to each device.

DHCP services exist for networks running Internet Protocol version 4 (IPv4), as well as version 6 (IPv6). The IPv6 version of the DHCP protocol is commonly called DHCPv6.

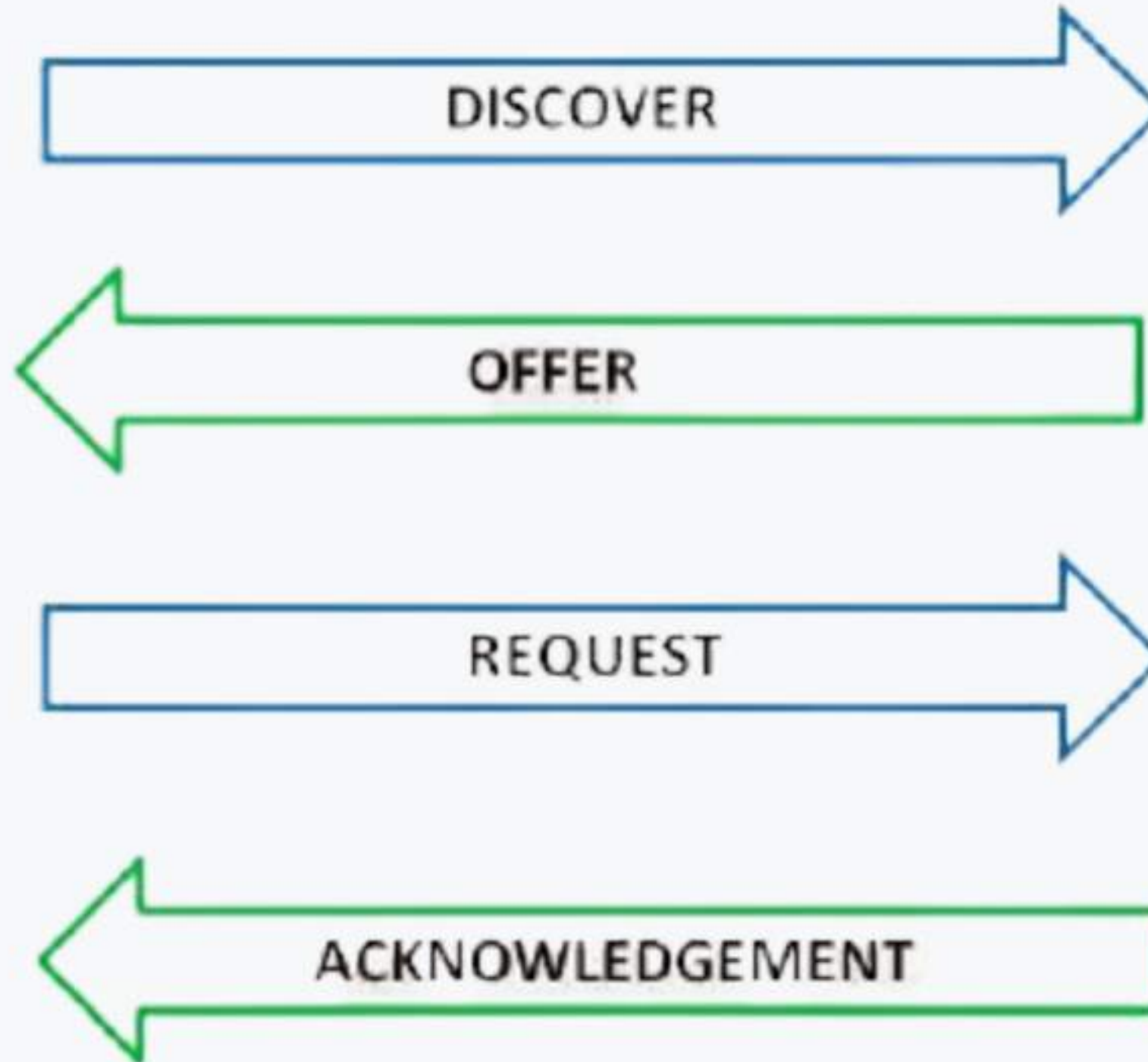
Dynamic Host Configuration Protocol



DHCP Working



DHCP Client



DHCP Server

Domain Name System



DNS stands for “Domain Name System”. It’s a system that lets you connect to websites by matching human-readable domain names (like wpbeginner.com) with the unique ID of the server where a website is stored.

Think of the DNS system as the internet’s phonebook. It lists domain names with their corresponding identifiers called IP addresses, instead of listing people’s names with their phone numbers. When a user enters a domain name like wpbeginner.com on their device, it looks up the IP address and connects them to the physical location where that website is stored.

he internet is possible because of a network of computers called servers. A server is a type of computer that’s dedicated to storing and delivering websites to other computers around the world.

A name server, sometimes spelled “nameserver”, is a special type of server that keeps all the DNS records of your domain name. Its job is to provide your DNS information to anyone requesting it.

Nameservers are typically managed by your domain name registrar or hosting provider.

Each nameserver has its own address and can store the records of many websites.

Domain Name System





SMTP Server

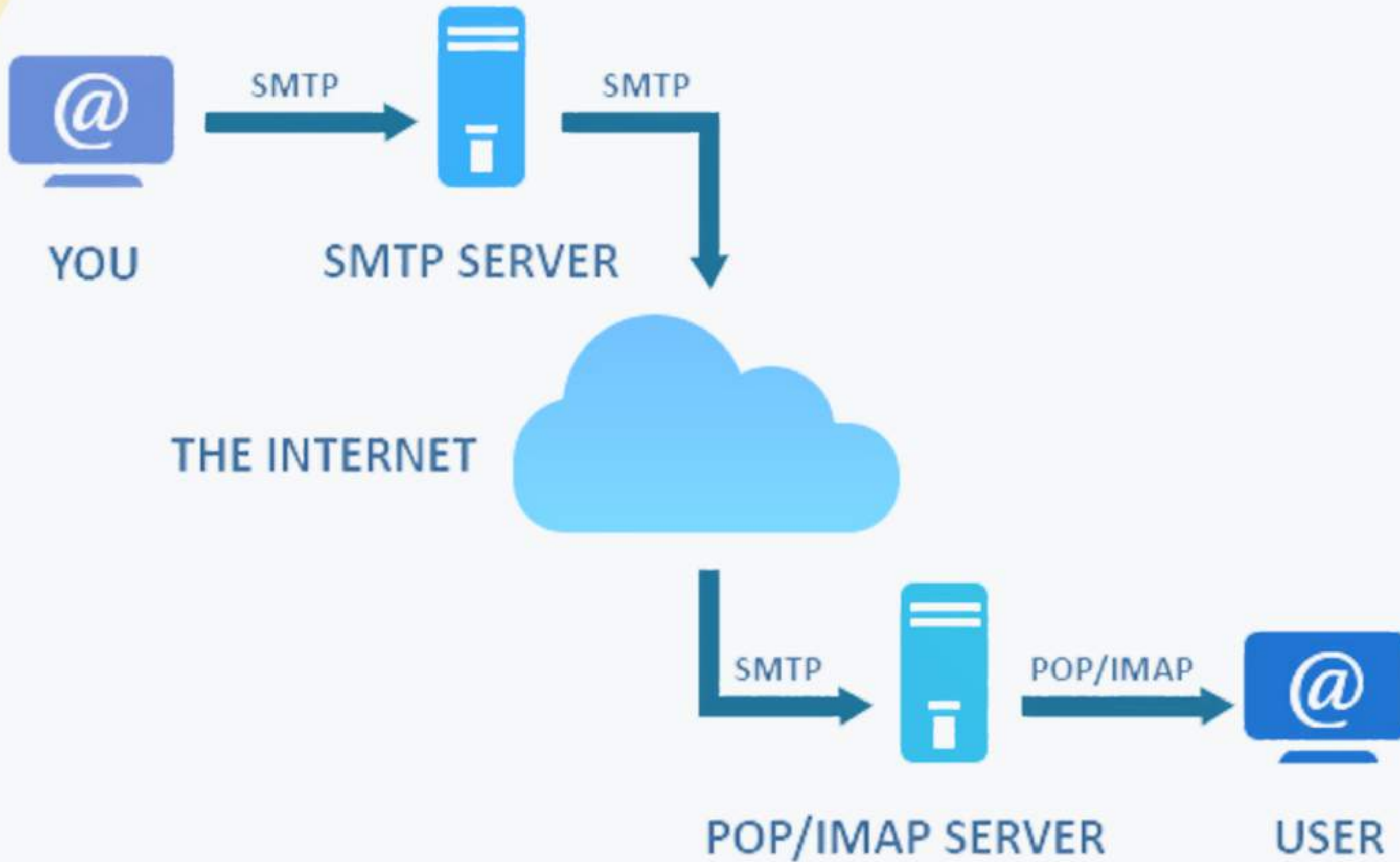
SMTP stands for Simple Mail Transfer Protocol, and it's an application used by mail servers to send, receive, and/or relay outgoing mail between email senders and receivers.

An SMTP email server will have an address (or addresses) that can be set by the mail client or application that you are using and is generally formatted as smtp.serveraddress.com. For example, Gmail's SMTP server host address is smtp.gmail.com, and Twilio SendGrid's is smtp.sendgrid.com. You can generally find your SMTP email server address in the account or settings section of your mail client.

When you send an email, the SMTP server processes your email, decides which server to send the message to, and relays the message to that server. The recipient's inbox service provider, such as Gmail or AOL, then downloads the message and places it in the recipient's inbox.

Port no. 25

SMTP Server



FTP Server



The computer which uses FTP to transfer data is called the FTP server. It stores and shares client data. Every day thousands of files on the Internet are transferred from one computer to another. Most of these files are transferred via the FTP server. It is an essential component of the FTP architecture.

The FTP

server stores the address of client files and creates a link to share these files. An FTP server requires a transfer control protocol network or internet protocol network to function. FTP server connection remains connected to FTP

clients all the time. It helps in transferring files from one user to another via the Internet. Any user can access the data stored in the FTP server, while some files can be accessed by special users.

SAMBA



Samba is an open-source software suite that lets Linux, Unix, and other non-Windows systems share files, printers, and other resources with Windows machines using the SMB/CIFS (Server Message Block/Common Internet File System) protocol.

Key points:

- **Purpose:** Enables interoperability between Windows and non-Windows systems.
- **Main uses:**
 - **Share directories and files over a network.**
 - **Act as a domain controller or member in a Windows Active Directory network.**
 - **Share printers with Windows clients.**
- **Compatibility:** Works with Windows, macOS, and other SMB-compatible clients.
- **Configuration:** Typically set up via smb.conf file, with tools like smbclient and testparm for testing.
- **Security:** Supports user-level authentication, encrypted connections, and integration with LDAP or Kerberos.

SAMBA



Step-by-Step Guide (Using Samba):

- **sudo apt install samba -y**
- **mkdir -p ~/shared_folder**
- **cp /path/to/yourfile.txt ~/shared_folder/**
You can replace /path/to/yourfile.txt with the actual path of your file.
- **sudo nano /etc/samba/smb.conf**
- **Scroll to the bottom and add this:**

[SharedFiles]

path = /home/your_username/shared_folder

available = yes

valid users = your_username

read only = no

browsable = yes

public = yes

- **writable = yes**

SAMBA



- **Replace your_username with your actual linux username.**
- **sudo smbpasswd -a your_username**
- **Enter a password (you'll use this to access the share from Parrot).**
- **sudo systemctl restart smbd**
- **sudo ufw allow samba**

Virtual private network



A virtual private network (VPN) extends a private network across a public network and enables users to send and receive data across shared or public networks as if their computing devices were directly connected to the private network. The benefits of a VPN include increases in functionality, security, and management of the private network. It provides access to resources inaccessible on the public network and is typically used for telecommuting workers. Encryption is common, although not an inherent part of a VPN connection.

A VPN is created by establishing a virtual point-to-point connection through the use of dedicated circuits or with tunneling protocols over existing networks. A VPN available from the public Internet can provide some of the benefits of a wide area network (WAN). From a user perspective, the resources available within the private network can be accessed remotely

Shell Scripting



The shells are usually interactive which means they receive commands as input through the users and run them. Although sometimes we routinely wish to run a set of commands, hence, we have to type within the commands all-time inside the terminal.

A shell script includes syntax similarly to other programming languages. When we have prior experience along with a programming language such as C/C++, Python, etc. It will be very easy to begin with it. The shell script combines the below components:

Functions

- Control flow: if, else, then, shell loops, case, etc.
- Shell commands: touch, pwd, echo, ls, cd, etc.
- Shell keywords: break, if, else, etc.